Tax Map

What a Lagos business owes under Nigeria's 2025 tax laws, with the section behind every line, checked by a second agent.

Tofarati Farinu · Team ISEGUN (solo) · Track 2: The Agent That Can Explain Why

Built with

Zetaris

Meterless

NVIDIA Nemotron

Cursor · Claude Code

The problem and who it is for

Nigeria's 2025 reform rewrote the tax code into four federal Acts, and obligations still arrive from three tiers of government. A shop owner in Ikeja has no single place to read what applies to her. A general chatbot answers from memory: no section, no way to check, and the law changed this year.

Who it is for

- Lagos SME owners and their accountants
- Junior revenue officers who field these questions daily
- Anyone building a compliance tool who needs a statute-grounded pattern

What they get

- What applies to them, and what they are exempt from
- The exact section behind each line, openable in full
- A second check that sets aside what the law does not support

Three tiers, one question

Federal
Nigeria Revenue Service
VAT, companies income tax, development levy

Lagos State
Lagos State Internal Revenue Service
Personal income tax, PAYE, premises registration

Ikeja LGA
Ikeja Local Government
Shop rates, tenement rate, signboard permits

The solution

One rule: the statute text is the source of truth, the model is the reader, and the code enforces it.

Researcher agent

Navigates a generated map of the Acts, browses Parts, reads sections in full, drafts obligations and exemptions

Critic agent

Judges every line against the quoted text and the Acts' own definitions; keeps, qualifies or sets aside

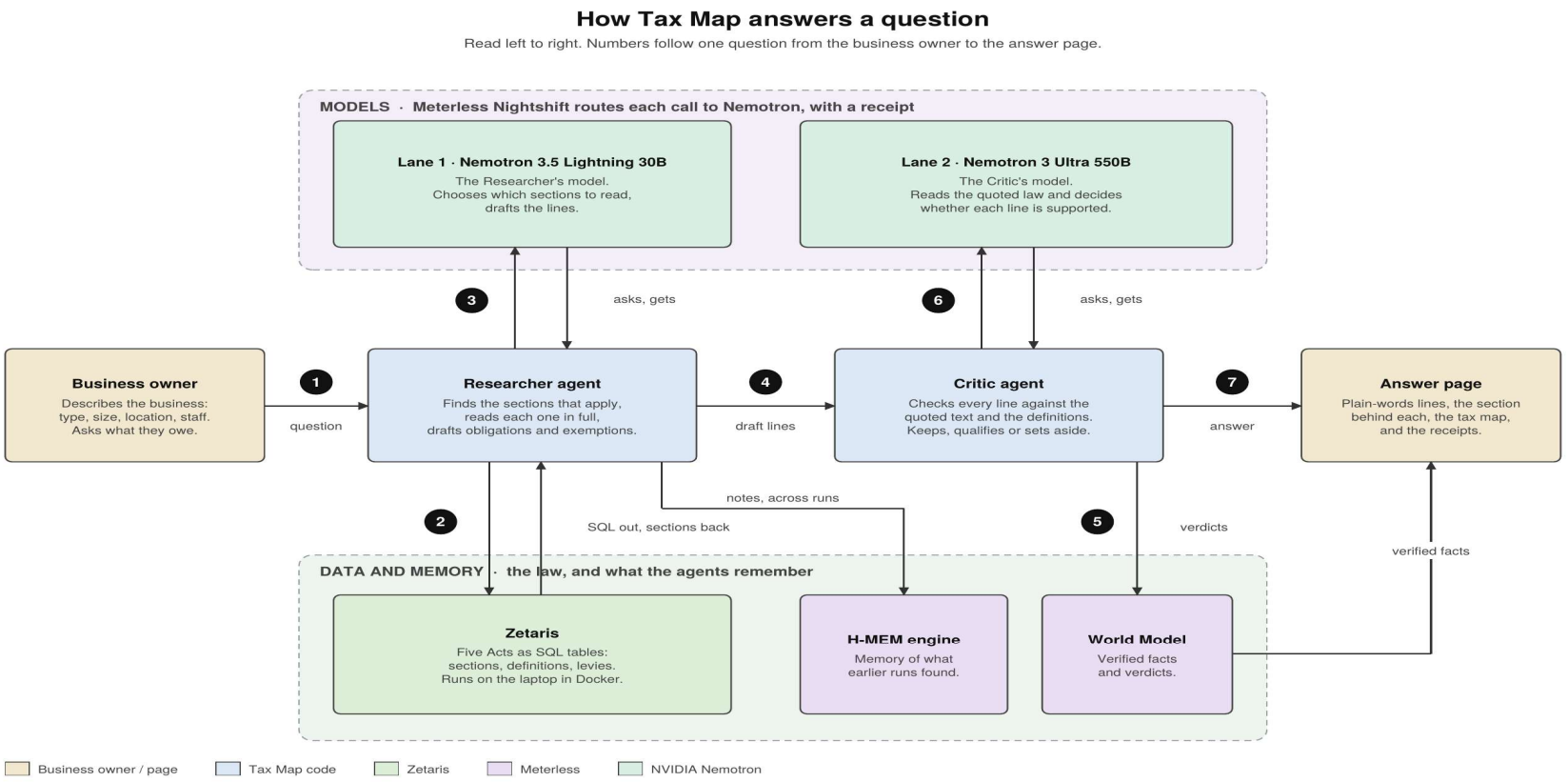
Answer page

Plain-words lines, the law behind each, a tax map by collecting authority, the receipts

What the code enforces, not the prompt

- Read before cite: a claim may cite only a section fetched in full
- Verified quotes: the Critic's quote must be verbatim in the cited text
- Definitions attached by code: 271 terms from the interpretation sections
- Claims-only coverage checklist across six obligation areas
- Levies: "may collect", never an amount; relevance to the business filtered
- Placement by collecting authority under NTAA s. 3

Agent workflow



Technical stack

App

- Node 20, TypeScript, Express
- Plain HTML, CSS and JS; no framework, no CDN
- Server-sent events stream each step
- 245 tests (vitest); none touch the network

Data

- pdfplumber parser: five Acts to section-level JSON
- Zetaris REST tables over pinned GitHub raw JSON
- Cached views: 40 s to 0.4 s per query

Models and memory

- Nemotron 3.5 Lightning 30B: Researcher
- Nemotron 3 Ultra 550B: Critic
- Meterless Nightshift 1.2.3 gateway, signed receipts
- Meterless H-MEM and World Model engines

454 + 39

statute sections and schedule items

271

defined terms attached automatically

7/7 · 8/8

engine conformance, H-MEM and World Model

245

tests passing in a clean clone

How I used Zetaris and Meterless

Zetaris: the data layer

- Five Acts parsed to JSON and served from pinned GitHub commits as Lightning REST tables
- Views taxmap.sections, taxmap.acts and taxmap.levies; the agent writes ordinary SQL
- Caching the views cut queries from 40 s to 0.4 s; the app re-caches on startup
- A judge can run the agent's exact SQL in the Zetaris SQL Editor

Meterless: the power layer and the engines

- Nightshift is the app's primary AI provider: every model call goes through it, lane-pinned, with a signed compute receipt
- H-MEM: tiered memory, hashed embeddings, forgetting curve, cross-run promotion, JSONL trust ledger (7/7)
- World Model: content-hashed graph, idempotent 10-stage ingest, claim context builder (8/8)
- Three bugs reported with repros; two fixed in Nightshift 1.2.3

Zetaris answers "what does the law say"; Meterless carries every model call and remembers what the agents found and verified.

What was different about how I used the sponsors

NVIDIA Nemotron

- Two agents, two Nemotron models, one job each: a fast drafter and a large judge
- Every receipt names the model; the demo shows a 550B judge striking a 30B draft
- No other model anywhere in the product path

Grounding by code

- The Critic's quote is checked against the statute text in code, so a hallucinated citation cannot survive
- Definitions are attached by code from the Acts' own interpretation sections
- Levies: "may collect", never an amount; placement by NTAA s. 3

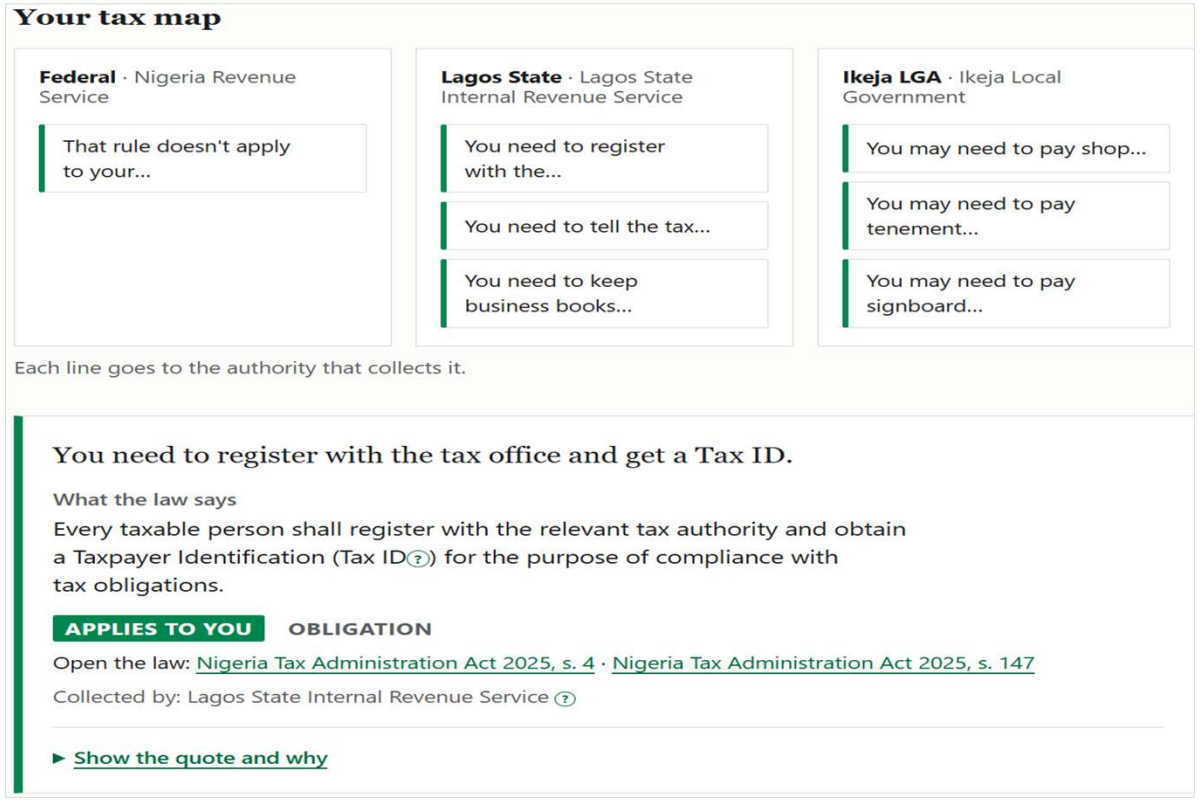
Cursor and Claude Code

- Built from written prompts; I reviewed every file and ran every test
- The organisers dropped the Cursor requirement on 22 Oct; both tools are disclosed in the README
- LICENSE, NOTICE and SOURCES with document hashes

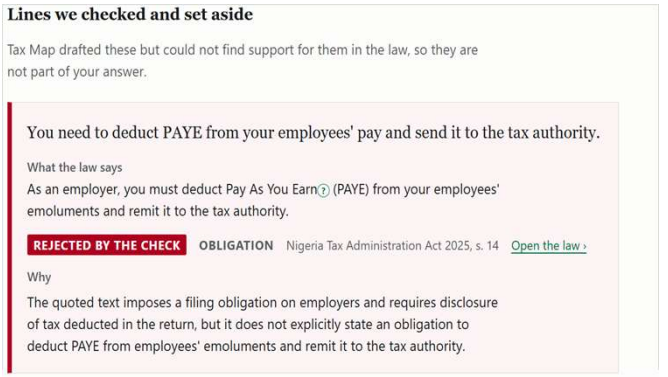
Where each sponsor sits in the product

Zetaris: the law | **Meterless: calls + memory** | **NVIDIA: both agents** | **Cursor · Claude Code: the build**

Product visuals



Results page: summary, tax map by authority, claim cards



A set-aside line with its reason and the disapplying quote



Receipts: every call on Nemotron, signed

Future enhancements

Lagos State and local law

Load the Lagos State Harmonised Taxes and Levies Law and Ikeja's bye-laws so the State and LGA columns cite local instruments, with rates

Schedules

Parse the Schedules of the 2025 Acts (rates and allocations) so the agents can cite them

Instant answers

Precompute answers for common business profiles from the World Model; serve them at once, dated, with a Run again button

More profiles

Companies, partnerships and other States; owner residence separate from business location

Known limits today

Federal Acts plus the 1998 allocation only; live runs take 6 to 12 minutes and vary between runs; sample mode replays a recorded run; not legal advice

Every line cites the law.
Every call has a receipt.

github.com/tman6290/tax-map · Tofarati Farinu · ISEGUN